

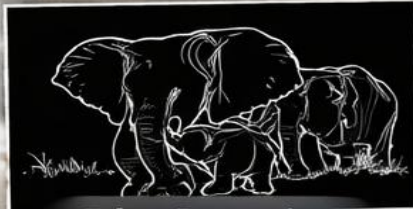
Task 2 Documentation

ADVANCED PHOTO EDITING SYSTEM

Using Python, OpenCV & NumPy



Thresholding



Edge Detection



Feature Detection



Panorama Stitching

Prepared by: _____

Nikita Anand

Practical Assessment Documentation

Name: Nikita Anand

Module Name: Computer Vision

Question: Create a photography image editing system using Python, OpenCV, and NumPy.

Aim

To implement an image editing system that can change brightness, remove noise, detect edges, find important features, and stitch multiple images together into a panorama using Python, OpenCV, and NumPy.

Requirements

- Computer System
- Python Software
- Jupyter Notebook / VS Code
- Libraries: OpenCV, NumPy

Steps

1. Import Libraries
 - `import cv2` `import numpy as np`
2. Load Images in Grayscale
 - `img1 = cv2.imread("elephant1.jpg", cv2.IMREAD_GRAYSCALE)`
 - `img2 = cv2.imread("elephant2.jpg", cv2.IMREAD_GRAYSCALE)`
3. Brightness Control (Thresholding)
`_, thresh = cv2.threshold(img1, 127, 255, cv2.THRESH_BINARY)`
4. Noise Removal (Gaussian Blur)
`gaussian = cv2.GaussianBlur(img1, (5, 5), 0)`
5. Edge Detection (Canny)
`edges = cv2.Canny(img1, 100, 200)`
6. Feature Detection (ORB)
`orb = cv2.ORB_create()`
`keypoints, descriptors = orb.detectAndCompute(img1, None)`
`img_features = cv2.drawKeypoints(img1, keypoints, None, color=(0,255,0))`

7. Panorama Stitching

```
stitcher = cv2.Stitcher_create()
(status, pano) = stitcher.stitch([img1, img2])
```

```
if status == cv2.STITCHER_OK:
    cv2.imshow("Panorama", pano)
    cv2.imwrite("panorama_result.jpg", pano)
else:
    print("Panorama stitching failed!")
```

8. Display Results

```
cv2.imshow("Original Grayscale", img1)
cv2.imshow("Threshold", thresh)
cv2.imshow("Gaussian Blur", gaussian)
cv2.imshow("Edges", edges)
cv2.imshow("Features", img_features)
```

```
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Screenshots

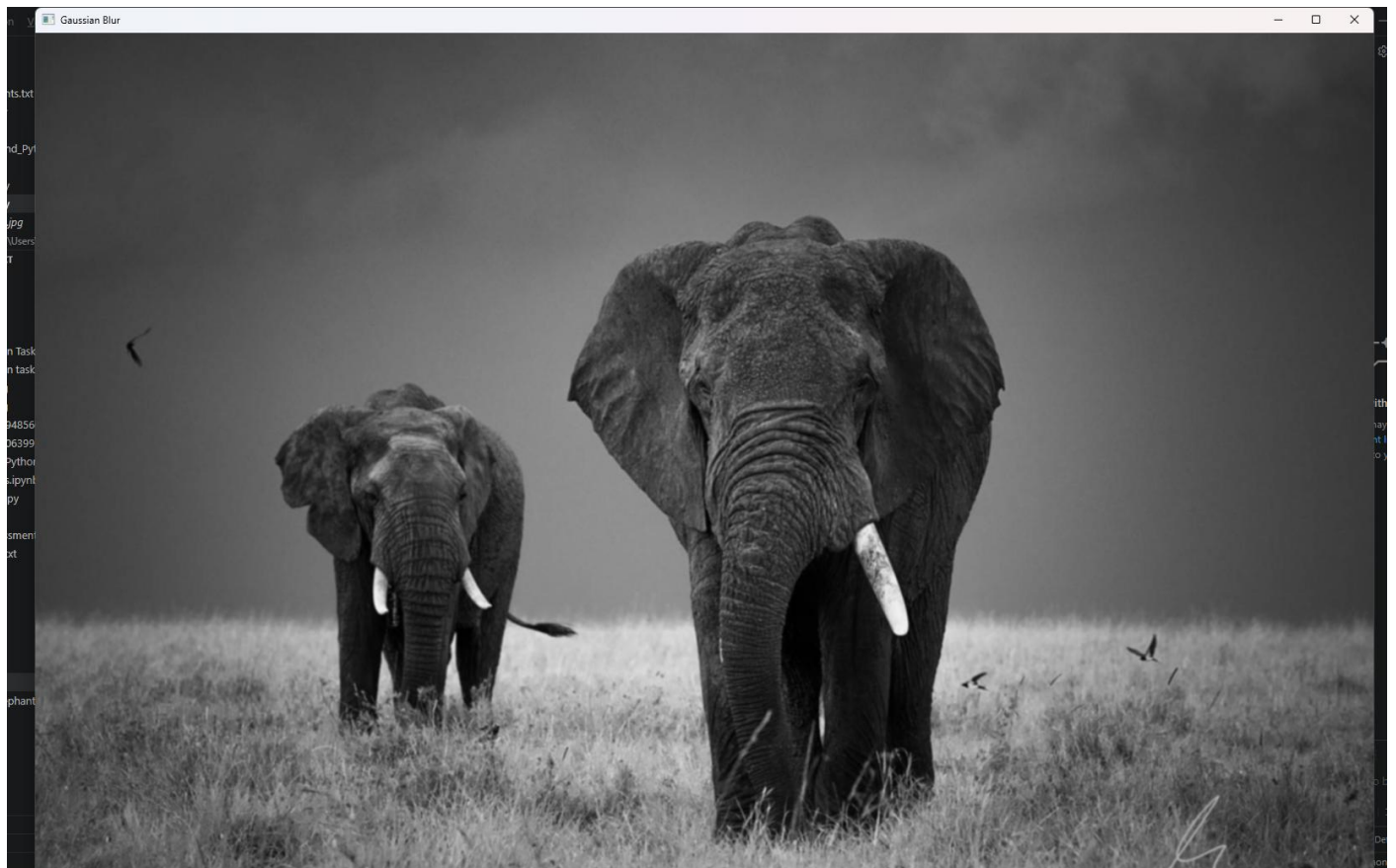
1. Original Grayscale Image



2. Thresholded Image



3. Gaussian Blur (Noise Removed)



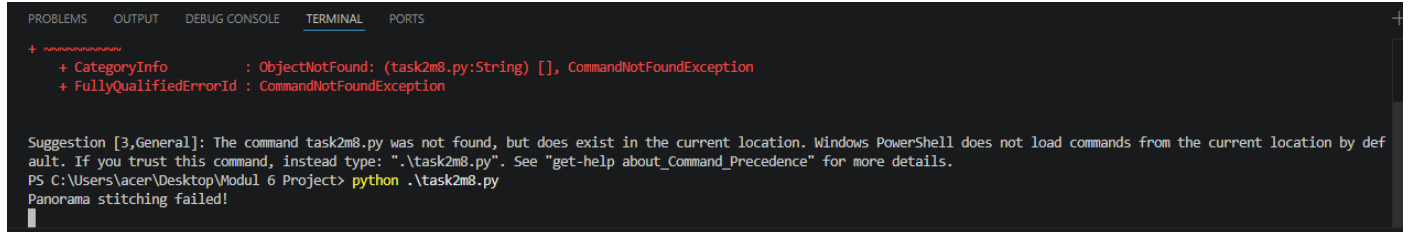
4. Edge Detection Result



5. Feature Detection (ORB Keypoints)



6. Panorama (stitched image)



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (task2m8.py:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

Suggestion [3,General]: The command task2m8.py was not found, but does exist in the current location. Windows PowerShell does not load commands from the current location by default. If you trust this command, instead type: ".\task2m8.py". See "get-help about_Command_Precedence" for more details.
PS C:\Users\acer\Desktop\Modul 6 Project> python .\task2m8.py
Panorama stitching failed!
```

Results

- Thresholding: Converted image to black & white based on brightness.
- Gaussian Blur: Removed noise and smoothed the image.
- Canny Edges: Detected object boundaries clearly.
- ORB Features: Highlighted important points in the image.
- Panorama: Combined two elephant photos into one wide image.

Conclusion

This project successfully demonstrates how OpenCV and NumPy can be used to build a photography image editing system. The system can adjust brightness, remove noise, detect edges, identify features, and stitch multiple images into a panorama. It provides a foundation for advanced photo editing applications similar to mobile photo editors.